A. SDM & Stacked SDM (SSDM): "Predict First, Assemble Later"

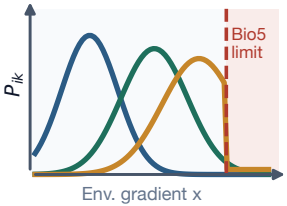
PO / PB records → individual SDMs → stacked assemblage (Ferrier & Guisan, 2006; Dubuis et al., 2011)

1. Presence-Only & Env.

Cell $\mathbf{x}_i$ (22 predictors)

Presence / BG $y_{ik} \in \{0, 1\}$

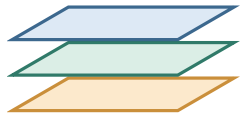
2. Individual Niches (S SDMs)



$P_{ik} = f_k(\mathbf{x}_i)$ (PB Logistic L/Q/H)

+ Realized Bio5/6 & water masks

Stack



Stack: $\hat{S}_i = \Sigma_k \hat{y}_{ik}$

3. Stack & Partition

α-Richness (ΔS) & Groups

Baselga β_{sim} vs. β_{sne}

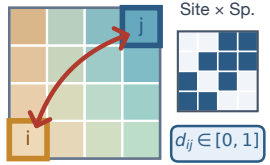
+ Strength: Retains taxon identity, realized heat attrition (β_{sne}), and α-richness (ΔS)

– Caveat: Independent stacking omits biotic coupling (addressed via overlap in Step 3) & rare spp.

B. Generalized Dissimilarity Model (GDM): "Assemble First, Predict Later"

Site-pair dissimilarity → monotonic I-splines & link → β turnover (Ferrier et al., 2007; Mokany et al., 2022)

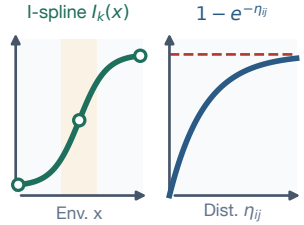
1. Site-Pair Dissimilarity



Site pairs (i, j): d_{ij}

$|\Delta I_k(x)|$ (22 vars) + Geo dist.

2. I-Splines & Exp. Link



Non-stationary 3-knot splines

+ Curvilinear $d_{ij} \to 1$ saturation

3. Beta-Diversity Outputs

Compositional Turnover
Temporal shift d_{GDM}

Disappearing Bioclimates
No future analogue (≥10% d)

Novel Bioclimates
No current analogue; rare taxa

+ Strength: Directly models β-diversity, nonlinear thresholds, & rare taxa in a single model

– Caveat: Space-for-time & lag; novelty extrap. (MESS/ExDet); sparse PO → $d_{ij} = 1$

C. National 1 km Climate × Land-Use Coupling: UK Observed GDM vs. Korean SSDM + Assemblage GDM

Great Britain Benchmark — Cooke et al. (2026, Nat. Commun.)

Structured Monitoring
1,002 plants, 56 butterflies
(UKBMS), 219 birds

→ 1 km GDM
31–56% deviance
β & bioclimates

× CRAFTY-GB Land Use
Brown et al. (2022)
× PREDICTS BII

→ SAR ($z = 0.25$)
Regional γ-diversity
extinction debt

South Korea Workflow — This Workshop (NES 3rd–5th)

Decadal Structured NES
Sparse PO: 45.5% $d_{ij} = 1$
(Raw GDM: 0.8% dev.)

→ 139-Spp. SSDM
22 vars + Dyna-CLUE
(100 m) + Bio5 masks

→ Assemblage GDM
73.7% dev. (smooth)
(11.8–14.8 °C)

+ Baselga Partition
β_{sne} (85% lowland)
vs. β_{sim} (60% high)